

Fundamental Diagram Modeling and Simulation of Mixed Traffic Flow Considering Functional Degradation

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ABSTRACT

Accurately modeling mixed traffic flow of Connected Automated Vehicles (CAVs) and Human-Driven Vehicles (HDVs) is crucial for efficiency and safety. This study proposes a microscopic car-following model that explicitly accounts for the degradation of CAVs into one-way-sensing "Degraded CAVs (DCAVs)" due to communication limitations or following HDVs, and derives an analytical fundamental diagram (FD) using the CAV penetration rate p as a control variable. The model captures steady-state flow–density–speed relationships, elucidating how p influences roadway capacity and critical speed through equivalent headway adjustments. SUMO-based simulations of an urban expressway merging scenario indicate that as p increases from 0 to 1, capacity rises from 1775 to 4421 veh/h, critical speed shifts upward, and optimal density slightly increases. Simulated and analytical FDs agree closely, with deviations within 1–2%. The results confirm that considering functional degradation prevents capacity overestimation and provides a quantitative basis for capacity evaluation and traffic management strategies.

INTRODUCTION

As CAV technologies progress from experiment to large-scale deployment, roadways are entering a prolonged transition in which human-driven vehicles (HDVs) and CAVs must share the same traffic stream—and learn to cooperate within it. Even a modest penetration of automation can reshape traffic dynamics: experiments show that an autonomous vehicle (AV) penetration of barely 5% can suppress stop-and-go waves, reducing fuel use and unnecessary braking (Stern et al., 2018). Meanwhile, vehicle-to-everything (V2X) communication offers a complementary path to coordination, enabling connected vehicles (CVs) to exchange data in real time, shorten headways, and dampen speed fluctuations—two fundamental levers for improving flow efficiency and service quality (Bansal & Kockelman, 2017).

On the modeling front, Talebpour and Mahmassani (2016) distinguished between HDVs, communication-only CVs, and automation-only AVs, and developed car-following models tailored to these technology assumptions. Their analysis of FD shapes and stability across penetration levels indicated that AVs are generally more effective than CVs at suppressing shock waves. The underlying reason lies in longitudinal control authority: although CVs can access richer external information via communication, longitudinal actuation often remains human-in-the-loop; by

contrast, AVs can autonomously regulate acceleration using adaptive cruise control (ACC), which inherently damps perturbation growth (Shladover, Su,&Lu, 2012). Extending this logic, CAVs—combining communication and automated control—can realize information and control coordination within platoons; cooperative adaptive cruise control (CACC) has been shown to substantially improve platoon stability and roadway efficiency (Nowakowski et al., 2015).

However, limited research has examined functional degradation when a CAV follows an HDV. Factors such as sensor occlusion, communication loss, or control anomalies can intermittently disrupt cooperation, causing deviations from the ideal state. Assuming perfect cooperation therefore risks overestimating stability and capacity gains. When a CAV reverts to ACC operation—functioning as a one-way perception degraded CAV (DCAV)—its behavior mimics an HDV, diminishing its ability to mitigate congestion. To address this issue, Zhang and Hu (2022) integrated a PATH-calibrated ACC model with the Gipps framework to construct a continuous cellular automaton model capturing CAV degradation. Their simulations showed that, at 60% CAV penetration, the proportion of congested vehicles decreased markedly and total capacity increased, implying a net benefit even under degradation. From a stability standpoint, Wang et al. (2019) found that string stability in heterogeneous traffic deteriorates significantly when CACC loses communication support and reverts to ACC. Likewise, recent modeling explicitly introducing CACC degradation indicates that communication instability reduces efficiency and heightens potential safety risks (Chen, Wu, & Liang, 2023).

Motivated by these insights, the primary objective of this paper is to establish a modeling framework that explicitly accounts for functional degradation in mixed traffic. Main contributions are threefold: (1) formalizing the functional degradation mechanism into a unified microscopic car-following model encompassing CAVs, DCAVs, and HDVs; (2) deriving an analytical FD based on steady-state analysis that elucidates how CAV penetration rates dictate roadway capacity through equivalent headway adjustments; and (3) validating the model’s accuracy and robustness using SUMO-based simulations of urban expressway merging segments, quantified by multiple goodness-of-fit metrics. Collectively, this work provides an interpretable quantitative basis for assessing the realistic benefits of CAV deployment, thereby rectifying the capacity overestimation risks inherent in ideal-cooperation models.

MIXED-TRAFFIC CAR-FOLLOWING MODEL

Car-Following Modes

We consider a single-lane, longitudinal mixed traffic stream composed of CAVs and HDVs. Let the CAV penetration rate be p and the HDV penetration rate be $1-p$. To capture possible communication loss and functional degradation in CAV→CAV interactions under high-density or high-speed conditions, we explicitly introduce a CAV–CAV cooperation availability $\rho_{cc}(k, v) \in [0, 1]$, where k denotes the traffic density and v the steady-state speed. The four representative follower-leader modes and their probability proportions are summarized in Tab. 1.

Table 1. Car-following modes and probability proportions in mixed traffic

No.	Interaction Type	Description	Probability Proportion
1	CAV→CAV (Cooperative)	Operates in fully cooperative mode with two-way perception.	$p_1 = p^2 \rho_{cc}(k, v)$
2	CAV→HDV (Degraded)	Behaves as a one-way sensing DCAV due to lack of cooperative data.	$p_2 = p^2(1 - \rho_{cc}(k, v)) + p(1 - p)$
3	HDV→CAV (Manual)	Follower is an HDV and cannot receive leader's cooperative data.	$p_3 = (1 - p)p$
4	HDV→HDV (Manual)	Follower is an HDV and behaves conventional human-driven following.	$p_4 = (1 - p)^2$

By construction, these four proportions satisfy the normalization $\sum p_i = 1$. To account for the adverse impacts of high-density occlusion, high-speed relative motion, and increased communication latency, $\rho_{cc}(k, v)$ is formulated as a calibratable logistic function, drawing on V2X connectivity modeling (Zhao et al., 2016):

$$\rho_{cc}(k, v) = \frac{1}{1 + \exp(-[\rho_0 - \beta_k k - \beta_v v])}, \quad \beta_k, \beta_v \geq 0 \quad (1)$$

where the sensitivity coefficients β_k and β_v quantify the response intensity of cooperative availability to changes in density and speed, respectively.

In the subsequent steady-state analysis, these modes are consolidated into three functional states: CAV, DCAV, and HDV. As shown in Fig. 1, these states are the key determinants of capacity, distinguished primarily by their equivalent time headways (e.g., $t_c = 0.6s$ for CAV vs. $t_d = 1.1s$ for DCAV).

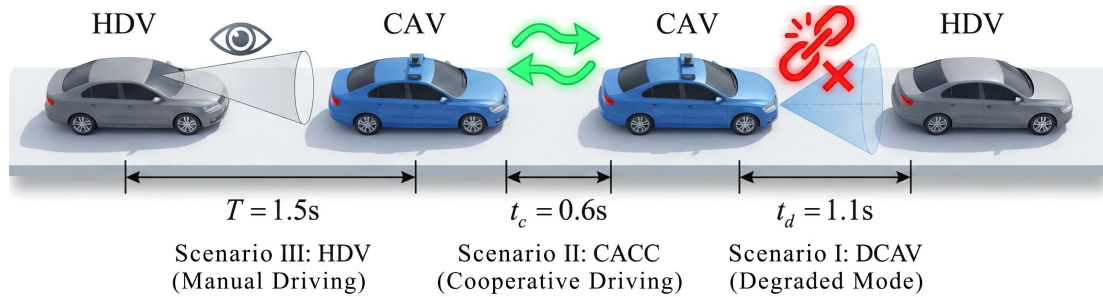


Figure 1. Vehicle interaction modes and functional degradation in mixed traffic

Modeling Vehicle Interactions with Functional Degradation

Equivalent Time Headway Mapping

To integrate functional degradation into steady-state spacing, communication and perception impairments (e.g., latency, noise, or disconnection) are mapped to an equivalent rise in safe time headway. Consider a follower–leader pair maintaining a steady speed v under sudden braking. Given the follower’s maximum comfortable deceleration, reaction delay τ , and perception error intensity σ relative to the leader’s deceleration b_ℓ , a sufficient condition for collision avoidance is:

$$h \geq l + s_0 + v\tau + \underbrace{\frac{v^2}{2b_f} - \frac{v^2}{2b_\ell}}_{\text{braking capability difference}} + \underbrace{\kappa_\sigma v \sigma}_{\text{uncertainty margin}} \quad (2)$$

where h is the bumper-to-bumper gap, l is vehicle length, s_0 is the minimum standstill gap, and $\kappa_\sigma \geq 0$ converts perception uncertainty into a time-headway reserve. Under CACC, $b_f \approx b_\ell$ and the quadratic term in Eq. (2) largely cancels; for DCAV/HDV cases it is non-negligible but its effect can be absorbed by the high-speed correction introduced in Eq. (4).

Since functional degradation directly alters the weighting of τ and σ , we parametrically adjust the physical time headway by incorporating the cooperative availability $\rho_{CC}(k, v)$ from Eq. (1) to derive the effective time headway:

$$T_{\text{eff}} = T_0 + \alpha\tau + \beta\sigma + \gamma(1 - \rho), \quad \alpha, \beta, \gamma \geq 0 \quad (3)$$

where T_0 is the baseline headway under ideal cooperation and $\rho \in [0, 1]$ denotes cooperation availability: for CAV $\rho = \rho_{CC}(k, v)$; for DCAV, $\rho = 0$ substituting T_{eff} into the steady-state desired spacing yields a unified form

$$s^*(v) = \underbrace{(l + s_0)}_{\text{minimum standstill gap}} + \underbrace{vT_{\text{eff}}}_{\text{time-headway-based safety gap}} + \underbrace{\frac{vT_h}{\sqrt{1 - (v/v_f)^\delta}}}_{\text{high-speed correction}} \quad (4)$$

where $T_h \geq 0$ and $\delta \in [2, 6]$ capture the rise of the safety margin at high speeds, and v_f is the free-flow speed.

CAV Car-following Model

For CAVs equipped with V2X, car-following behavior is represented by CACC, which adheres to a constant desired time headway principle (Milanés et al., 2014). The dynamics are given by

$$a_n(t) = k_1 a_{n-1}(t) + k_2 [x_{n-1}(t) - x_n(t) - t_c v_n(t) - l - s_0] + k_3 [v_{n-1}(t) - v_n(t)] \quad (5)$$

where a_n , x_n , and v_n denote the acceleration, position, and speed of vehicle n ; t_c is the cooperative time headway, set to 0.6s (Qin et al., 2017); l is vehicle length, set to 5m; and s_0 is the minimum standstill gap, set to 2m. Typical controller gains following Milanés et al. (2014) are $k_1 = 1.0$, $k_2 = 0.2$, and $k_3 = 3.0$.

Under steady-state ($\dot{v} = 0, \Delta v = 0$), spacing implied by Eqs. (2)-(3) becomes

$$h_c(v) = l + s_0 + vT_{\text{eff}}^{(c)}, \quad T_{\text{eff}}^{(c)} = T_{0,c} + \alpha\tau_c + \beta\sigma_c + \gamma[1 - \rho_{CC}(k, v)]$$

By integrating leader acceleration, relative spacing, and relative speed, the CACC formulation ensures efficient and stable cooperative following.

DCAV Car-following Model

When a CAV follows an HDV, or when a CAV-CAV pair experiences communication loss, cooperation degrades into one-way-sensing DCAV operation. Experiments from the PATH program and subsequent studies indicate that an ACC-type model effectively characterizes DCAV following (Xie et al., 2019):

$$a_n(t) = j_1 [x_{n-1}(t) - x_n(t) - t_d v_n(t) - l - s_0] + j_2 [v_{n-1}(t) - v_n(t)] \quad (6)$$

where the degraded headway parameter is $t_d = 1.1\text{s}$ (Qin et al., 2017). Following Xie et al. (2019), representative parameter values are $j_1 = 0.23\text{s}^{-2}$ and $j_2 = 0.07\text{s}^{-2}$. Under steady state, Eqs. (2)-(3) yields

$$h_d(v) = l + s_0 + vT_{\text{eff}}^{(d)}, \quad T_{\text{eff}}^{(d)} = T_{0,d} + \alpha\tau_d + \beta\sigma_d \quad (\rho = 0)$$

Unlike CACC, the DCAV model lacks acceleration feedforward, which inevitably reduces stability and roadway capacity compared to ideal cooperation.

HDV Car-following Model

The car-following behavior of HDVs is modeled using the Intelligent Driver Model (IDM) (Xie et al., 2019; Jia et al., 2019). Its formulation is expressed as

$$a_n(t) = a \left[1 - \left(\frac{v_n(t)}{v_f} \right)^4 - \left(\frac{s_0 + v_n(t)T + \frac{v_n(t)\Delta v_n(t)}{2\sqrt{ab}}}{h(t) - l} \right)^2 \right] \quad (7)$$

where $v_n(t)$ is the vehicle's speed, $\Delta v_n(t) = v_n(t) - v_{n-1}(t)$ is the relative speed, and $h(t) = x_{n-1}(t) - x_n(t)$ is the headway. Typical parameter values are $a = 2\text{m/s}^2$, $b = 2\text{m/s}^2$, $v_f = 33.3\text{m/s}$, and $T = 1.5\text{s}$.

The steady-state solution of the IDM inherently captures the high-speed asymptotic behavior of HDVs, providing a realistic upper bound for safe spacing and

acceleration at elevated velocities. Notably, the steady-state closed-form solution of the IDM exhibits a nonlinear increase in safety spacing with speed, providing the theoretical basis for the high-speed correction term in Eq. (4).

FUNDAMENTAL DIAGRAM FOR MIXED TRAFFIC

Building on the microscopic steady-state analysis, this section derives the mixed-flow FD to elucidate macroscopic dynamics under different penetration levels, followed by the corresponding solution procedures and capacity estimation methods.

Steady-state and Expected Headways

In steady state, $\dot{v} = 0$ and $\Delta v = 0$. Using the time-headway mapping in Eqs. (2)-(3), the steady-state spacings are

$$h_c(v) = l + s_0 + vT_{\text{eff}}^{(c)}, \quad h_d(v) = l + s_0 + vT_{\text{eff}}^{(d)}, \quad h_e(v) = l + \frac{s_0 + vT}{\sqrt{1 - (v/v_f)^4}} \quad (8)$$

Combining these with the mode proportions mode 1–4 from Tab.1 yields the expected steady-state headway of the mixed stream:

$$H(v; k, p) = p^2 \rho_{cc}(k, v) h_c(v) + [p^2 (1 - \rho_{cc}(k, v)) + p(1 - p)] h_d(v) + (1 - p) h_e(v) \quad (9)$$

When $\rho_{cc} \equiv 1$ and $T_{\text{eff}}^{(c)} = t_c, T_{\text{eff}}^{(d)} = t_d$, (9) reduces to the familiar convex combination special case.

Analytical Fundamental Diagram

To incorporate functional degradation consistently at the macroscopic level, note that the expected headway H in (9) couples with density k and speed v through $\rho_{cc}(k, v)$, leading to an implicit density–speed–flow relationship. Using the spacing–density identity gives the fixed-point equation

$$k(v; k, p) = \frac{1000}{H(v; k, p)} \quad (10)$$

where the factor 1000 ensures k is expressed in veh/km. Combining this with the flow identity $Q = 3.6kv$ yields the implicit flow–speed relation in veh/h:

$$Q(v; k, p) = \frac{3600v}{H(v; k, p)} \quad (11)$$

Capacity and Critical State

Because H depends explicitly on the cooperation availability $\rho_{cc}(k, v)$, equations (8)–(11) form an implicit nonlinear system for the density–flow pair (k, Q) . To obtain stable solutions, we apply a fixed-point iteration. For a given penetration

rate p and steady speed v , initialize $k^{(0)} \in (0, k_{\max})$ and iterate

$$H^{(m)} = H(v; k^{(m)}, p), \quad k^{(m+1)} = \frac{1000}{H^{(m)}}, \quad Q^{(m+1)} = \frac{3600v}{H^{(m)}}$$

until the convergence tests $|k^{(m+1)} - k^{(m)}| < \varepsilon_k$ and $|Q^{(m+1)} - Q^{(m)}| < \varepsilon_Q$ are satisfied, where ε_k and ε_Q are the density and flow tolerances, respectively. This procedure yields consistent steady-state density and flow for any (v, p) and recovers a continuous, unimodal flow–speed curve. The $Q - k$, $Q - v$, and $k - v$ relationship curves under different penetration rates are illustrated in Fig. 2.

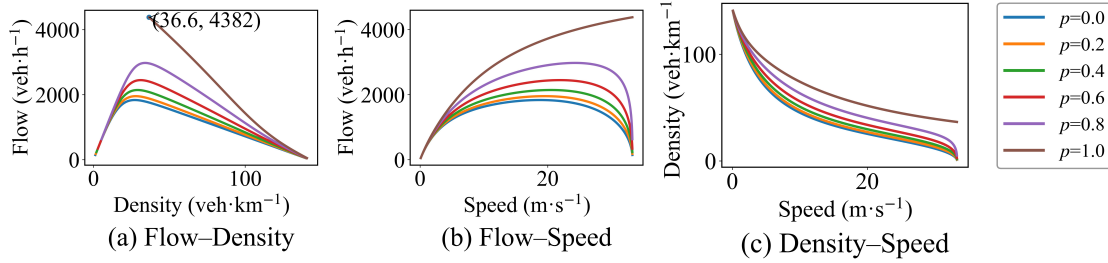


Figure 2. Flow, density, and speed relationships of mixed traffic

On this basis, the capacity of the mixed stream is defined as the maximum of the flow–speed curve over the feasible speed range $(0, v_f)$:

$$Q_{\text{cap}}(p) = \max_{v \in (0, v_f)}, Q(v; p) v_{\text{cr}}(p) \in \arg \max_v Q(v; p), k_{\text{opt}}(p) = k(v_{\text{cr}}(p); p)$$

Here, $Q_{\text{cap}}(p)$ is the maximum achievable flow at p , $v_{\text{cr}}(p)$ is the critical speed, and $k_{\text{opt}}(p)$ is the corresponding optimal density. Since $Q(v; p)$ is continuous in v and has a unique interior maximizer on $(0, v_f)$, $v_{\text{cr}}(p)$ can be obtained via a one-dimensional search. Tab. 2 summarized the computed Q_{cap} , v_{cr} and k_{opt} .

Table 2. Traffic capacity indicators and sensitivity analysis of mixed traffic flow under different penetration rates

p	Max flow (veh/h)	Optimal density (veh/km)	Critical speed (m/s)	Marginal growth (%)	Cumulative growth (%)
0.0	1774.9	30.2	16.3	—	—
0.2	1888.5	30.3	17.3	6.4	6.4
0.4	2070.0	31.9	18.0	9.6	16.6
0.6	2361.8	33.9	19.3	14.1	33.1
0.8	2874.0	36.8	21.7	21.7	61.9
1.0	4421.1	37.6	32.7	53.8	149.1

Tab. 2 revealing a nonlinear capacity trend. Marginal growth is constrained ($<10\%$) at low penetration ($p \leq 0.4$) by large HDV headways but surges (20% – 54%) when $p \geq 0.6$, eventually increasing capacity by 2.5 times. This pattern reflects the CACC weighting mechanism in Eq. (9): higher p compresses steady-state headways and elevates critical speed. These results confirm the model's capability to capture flow dynamics and establish a theoretical benchmark for simulation validation.

SIMULATION VALIDATION

We establish simulation environments using the microscopic simulator SUMO. Numerical experiments are conducted to analyze mixed traffic dynamics across varying CAV penetration. The primary objectives are to verify: (1) the consistency of simulated $Q-k$, $Q-v$, and $k-v$ relationships with theoretical predictions; and (2) the agreement of capacity sensitivity to CAV penetration with analytical derivations.

Scenario and Setup

The simulation replicates a real-world urban expressway merging section, with geometric configuration and operational characteristics referenced from the Ubiquitous Traffic Eyes (UTE) dataset (Feng, 2024), as shown in Fig. 3. The section covers lanes 6–10, with a total length of 362m. Lane Area Detectors (E2) are deployed every 10 m in each lane along the segment to collect traffic flow, density, and speed data, and the Erdmann model is adopted for lateral lane-changing behavior.

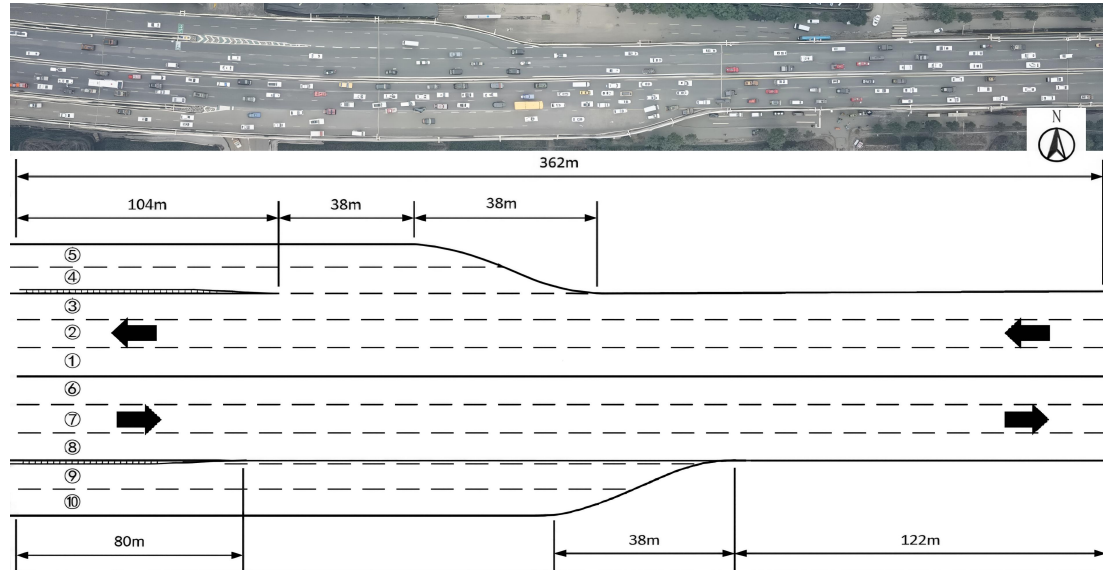


Figure 3. Aerial view and geometric layout of the expressway segment

To ensure the comparability between the analytical fundamental diagram model and the simulation results, the key parameters governing longitudinal car-following and control are inherited from the previous sections. On this basis, to capture short-term uncertainties arising from real driving behavior and onboard perception and control, small random perturbations are superimposed—via TraCI scripts—on the acceleration, desired time headway, minimum gap, and reaction delay parameters of HDVs and DCAVs. All perturbations are modeled using zero-mean

truncated normal distributions, whose truncation bounds are inferred from residual samples of the UTE trajectory dataset, as summarized in Tab. 3.

Table 3. Trajectory-Data-Calibrated Random Disturbance Settings

Perturbation Item	Distribution Form	Value Range
Acceleration ϵ_a	$N(0, \sigma_a^2)$	$[-0.30, 0.40]$ m/s ²
Desired time gap ϵ_T	$N(0, \sigma_T^2)$	$[-0.15, 0.15]$ s
Minimum gap ϵ_{s0}	$N(0, \sigma_{s0}^2)$	$[-0.50, 0.50]$ m
Reaction delay ϵ_τ	$N(0, \sigma_\tau^2)$	$[-0.20, 0.20]$ s

The CAV penetration rate is set as $p \in (0.0, 0.2, 0.4, 0.6, 0.8, 1.0)$, with the remaining vehicles being HDVs. A transition from CAV to DCAV is triggered when the following coupled conditions are satisfied:

(a) The target vehicle is a CAV, and its immediate preceding vehicle in the same lane is an HDV, with the headway between the two vehicles not exceeding the effective distance threshold $R = 30$ m;

(b) No communication signal feedback from the preceding vehicle is received for two consecutive sampling intervals, where the sampling interval is 5 s.

Results Analysis

Simulation results under varying CAV penetration rates are compared with the analytical FD curves in Fig. 4. This section evaluates model fidelity by examining goodness-of-fit metrics across different values of p , thereby assessing the accuracy and consistency of the analytical FD relative to the simulated scatter data.

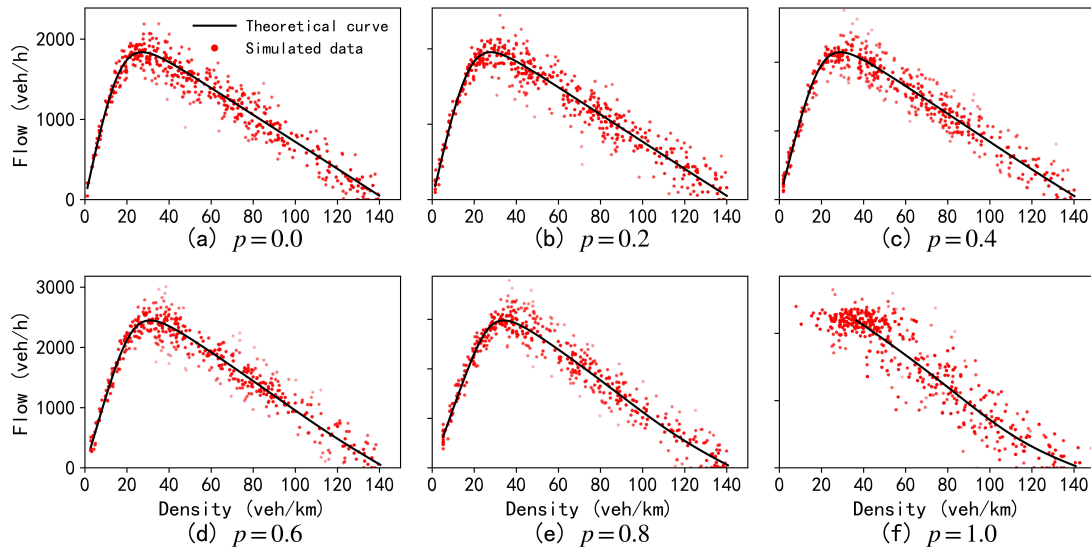


Figure 4. Comparison between simulation results and analytical FD model

We evaluate the agreement between simulated scatter points and the analytical FD using four metrics: root mean square error (RMSE), coefficient of determination (R^2), peak errors (ε^Q and ε^v), and bandwidth coverage ratio (BCR). For a given penetration rate p , suppose that after the warm-up period we collect N samples (k_i, Q_i^{sim}) , and the analytical curve yields $Q_i^{\text{th}} = Q(k_i; p)$. The metrics are defined as

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (Q_i^{\text{sim}} - Q_i^{\text{th}})^2}, \quad R^2 = 1 - \frac{\sum_{i=1}^N (Q_i^{\text{sim}} - Q_i^{\text{th}})^2}{\sum_{i=1}^N (Q_i^{\text{sim}} - \bar{Q}^{\text{sim}})^2}$$

Let the capacity peak obtained from theory and simulation be $(Q_{\text{cap}}^{\text{th}}, v_{\text{cr}}^{\text{th}})$ and $(Q_{\text{cap}}^{\text{sim}}, v_{\text{cr}}^{\text{sim}})$, respectively. The relative peak errors are

$$\varepsilon^Q = \frac{|Q_{\text{cap}}^{\text{sim}} - Q_{\text{cap}}^{\text{th}}|}{Q_{\text{cap}}^{\text{th}}} \times 100\%, \quad \varepsilon^v = \frac{|v_{\text{cr}}^{\text{sim}} - v_{\text{cr}}^{\text{th}}|}{v_{\text{cr}}^{\text{th}}} \times 100\%$$

Using a tolerance band $\tau = 5\%$, the bandwidth coverage ratio is

$$\text{BCR}(\tau) = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left(\left| \frac{Q_i^{\text{sim}} - Q_i^{\text{th}}}{Q_i^{\text{th}}} \right| \leq \tau \right) \times 100\%$$

To mitigate the influence of extreme transient congestion, observations with Q above the 99th percentile (less than 1% of samples) are excluded prior to computing RMSE, R^2 , ε^Q , ε^v , and BCR. The results are summarized in Tab. 4.

Table 4. Goodness-of-fit metrics between simulated data and analytical FD at different p

p	RMSE	R^2	$\varepsilon_Q(\%)$	(%)	BCR@ $\pm 5\%$ (%)
0.0	142.6	0.963	2.1	2.4	86.3
0.2	131.4	0.969	1.9	2.2	88.7
0.4	118.1	0.975	1.6	1.9	90.9
0.6	103.7	0.981	1.4	1.7	92.8
0.8	88.9	0.986	1.2	1.5	95.1
1.0	127.5	0.972	1.8	2.1	93.0

Overall, as p increases, RMSE decreases while both R^2 and BCR increase, indicating improved explanatory power and coverage of the analytical FD. The peak

errors ε^Q and ε^v remain low (1–2%), showing that capacity and critical-speed estimates are tightly matched. At $p = 1.0$, non-equilibrium disturbances in the high-density regime cause a slight uptick in RMSE and peak errors and a modest decline in BCR; nevertheless, the fit remains strong ($R^2 > 0.97$, BCR > 90%). This trend is consistent with the mechanism discussed earlier: higher levels of cooperation shift steady-state spacing toward a near-constant-headway regime, making macroscopic curves smoother and more predictable.

CONCLUSION

This study develops an analytical FD framework that explicitly accounts for CAV functional degradation in mixed traffic. By incorporating a density-speed dependent cooperation-availability function, it unifies ideal CACC, degraded ACC, and manual IDM behaviors within a single formulation. Steady-state analysis further shows that the mixed-traffic headway can be represented as a convex combination of distinct following modes, providing a mechanistic explanation for the accelerated capacity growth when the CAV penetration rate exceeds ~ 0.6 . Validation using calibrated SUMO simulations in a real-world urban expressway merging segment shows close agreement with the analytical FD, achieving deviations below 2% for both capacity and critical speed across tested penetration rates, and reproducing the nonlinear FD shape over the free-flow and near-critical regimes. By bridging microscopic car-following dynamics and macroscopic traffic performance, the framework offers a robust quantitative basis for capacity evaluation and phased CAV deployment planning. Future work will focus on refining the degradation mechanism under realistic communication conditions and strengthening data-driven calibration across diverse operational regimes to further enhance predictive reliability.

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